Classification of the BUSI Dataset

Breast Ultrasound Image Classification using Deep Learning

1. Introduction

Breast cancer is among the most prevalent malignancies globally; early detection is critical for improving patient outcomes. Ultrasound imaging is widely used due to its non-invasive nature and real-time capability, yet manual interpretation is time-consuming and subject to inter-observer variability. This project applies a custom Convolutional Neural Network (CNN) to automatically classify images from the BUSI dataset into three clinically meaningful categories: benign, malignant, and normal.

2. Dataset & Split

The BUSI dataset (Cairo University) contains 1,578 images with notable class imbalance. An 80/20 stratified split yields 1,262 training and 316 test images.

Class	Samples	Proportion	Clinical Meaning
Benign	~487	30.9 %	Non-cancerous growth
Malignant	~210	13.3 %	Cancerous tumour
Normal	~133	8.4 %	No abnormality
Total	1,578	100 %	—

3. Methodology

3.1 Preprocessing

Images are resized to 150×150 px, converted to single-channel grayscale (reducing memory without losing diagnostic information), reshaped to (150,150,1), and pixel values normalised to [0,1].

3.2 Model Architecture

A custom Sequential CNN with four convolutional blocks captures hierarchical features (~6.94 M trainable parameters):

Layers	Configuration	Output Shape
Conv Block 1–4	Conv2D (32→64→128→256, 3×3, ReLU) + MaxPool 2×2	7×7×256 after block 4
Dense Layers	Flatten → Dense(512,ReLU) → Dropout(0.5) → Dense(256,ReLU) → Dropout(0.5)	256
Output	Dense(3, Softmax)	(3,)

3.3 Training Configuration

Hyperparameter	Value
Optimiser	Adam (β_1=0.9, β_2=0.999)
Loss	Categorical Cross-Entropy
Batch / Epochs	32 / 20
Validation Split	10 % of training set

4. Results & Evaluation

Metric	Training	Testing
Accuracy	98.49 %	83.23 %
Loss	0.0805	0.9758

The 15 % gap between training and test accuracy indicates overfitting — a common challenge on small medical imaging datasets. Per-class performance:

Class	Precision	Recall	F1-Score	Support
Benign	0.85	0.89	0.87	179
Malignant	0.74	0.73	0.73	84
Normal	0.91	0.79	0.85	53
Macro Avg	0.84	0.80	0.82	316

Confusion matrix highlights: benign (160/179 correct), normal (42/53 correct), malignant (61/84 correct). The most critical error — 16 malignant cases misclassified as benign — underscores the need for improved recall on the malignant class.

5. Challenges

- Class Imbalance: Benign dominates (~49 %); no oversampling or class-weighted loss was applied.
- Overfitting: Training/test gap of ~15 %; Dropout provided only partial regularisation.
- Small Dataset: 1,578 images is limited for training a deep CNN from scratch.
- No Data Augmentation: No flips, rotations, or brightness jitter were applied.
- Image Quality: Speckle noise, low contrast, and acquisition artefacts increase intra-class variation.

6. Conclusion & Future Work

The custom CNN achieves 83.23 % test accuracy and a macro F1 of 0.82, establishing a functional baseline. Future improvements include: (1) Transfer Learning (ResNet50, EfficientNet), (2) Data Augmentation, (3) Class-Balanced Loss, (4) Segmentation-guided Classification, (5) CLAHE / speckle-reduction preprocessing, and (6) Ensemble Methods.

References

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